



RS6xxx Radar Radome Design Guidelines

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1. The impact of radome on mmWave radar

1.1. Design Elements of Radome

As an important component of mmWave radar, the performance of the radome directly affects key indicators such as the measurement range, angular error, and anti-interference capability of the mmWave radar. The performance of the radome mainly includes structural performance and electrical performance. Structural performance mainly includes the appearance and structural strength of the radome. The appearance mainly involves aesthetics and concealment. The structural strength determines whether it can protect the antenna in harsh environments. Electrical performance mainly ensures that the radome has excellent electromagnetic wave transmission characteristics, which means it cannot block the antenna's radiation signal or have a serious impact on important indicators such as antenna gain and beamwidth. Therefore, both structural and electrical performance factors need to be fully considered when designing the radome.

1.2. Influence Factor of Radome

Signal attenuation: Depending on the type of radome material, its dielectric constant and dielectric loss vary, leading to different absorption or reflection of electromagnetic wave signals. This results in inconsistent attenuation of electromagnetic wave signals, which in turn affects the gain of the antenna and ultimately impacts the detection range and accuracy of the radar.

Scattering: In the actual radar environment, irregularities in the radome structure can cause scattering. When electromagnetic wave signals encounter the radome, they undergo reflection, refraction, and diffraction, leading to attenuation of the electromagnetic wave signals and the generation of multipath effects, thereby affecting the signal quality of the radar.

Phase delay: The dielectric constant of the radome differs from that of air. When electromagnetic wave signals pass through the radome, their phase will change. Due to the difference in path lengths, radomes with uneven thicknesses will produce varying degrees of phase delay, thereby affecting the measurement accuracy of radar.

Polarization effect: The radome may alter the polarization state of the incident electromagnetic wave signal. Especially when the radome is not transparent to electromagnetic wave signals, the polarization shift can lead to a decrease in signal reception strength, thereby affecting the radar's recognition capability.

Blind zone: The design and installation position of the radome can affect the blind zone of the radar, which refers to the area that the radar cannot detect.

In summary, radomes play a crucial role in achieving excellent radar performance. During the design process of radomes, it is necessary to consider their impact on the radar and select suitable materials, structures, and installation locations to ensure the detection performance of the radar.

2. Design principles of mmWave radar radome

2.1. Radome wall Thickness

The wall thickness of the radome is typically determined based on factors such as the operating frequency of the radar, the material of the radome, and the manufacturing process. In practical applications, the wall thickness of the radome not only determines the strength to protect radar components but also has a certain impact on the performance of the radar. Therefore, when designing the radome, it is necessary to comprehensively consider both the performance of the radar and the practical application requirements, and select an appropriate thickness.

The ideal radome wall thickness is an integer multiple of half the wavelength of the medium corresponding to the actual operating frequency of the radar, which ensures that the radome becomes "almost transparent" within the expected frequency range. Therefore, the thickness of the radome can be calculated using the following formula:

$$T = n * \frac{\lambda_m}{2} \quad (1)$$

$$\lambda_m = \frac{c}{f_0 * \sqrt{\epsilon_r}} \quad (2)$$

Where T is the thickness of the radome wall, n is a positive integer, λ_m is the wavelength in the medium, c is the speed of light in vacuum, f_0 is the center frequency of the radar operation, ϵ_r and ϵ_r is the relative permittivity of the radome material.

2.2. Distance D from the Radar to the Radome

The optimal distance from the antenna to the inner surface of the radome helps to minimize the reflection impact caused by the radome. When the electromagnetic wave reflected back to the antenna is in phase with the transmitted wave, this impact is minimized. Therefore, the ideal distance is an integer multiple of half a wavelength in air, which can be calculated using the following formula:

$$D = n * \frac{\lambda_0}{2} \quad (3)$$

$$\lambda_0 = \frac{c}{f_0} \quad (4)$$

Where D represents the distance from the antenna to the inner surface of the radome, n is a positive integer, λ_0 is the wavelength in air, c is the speed of light in vacuum, f_0 and f is the center frequency of radar operation.

2.3. Selection of Radome Materials

Materials with higher dielectric constant and higher loss tend to cause enhanced reflection and increased loss of electromagnetic wave signals. Therefore, materials with lower dielectric constant and low loss should be chosen as much as possible, and their electrical parameters should not change significantly with temperature and frequency. Generally, materials for radar radomes should ideally have a dielectric constant (DK) lower than 10 and a loss tangent (DF) lower than 0.01. When designing radar radomes, it is important to avoid using coatings containing metallic materials, as this is crucial for achieving excellent radar performance.

In addition, the materials used should be essentially homogeneous to avoid any additional DK boundaries within the radar radome. The design aims to achieve a solid radome wall, with no bubbles or other material fragments inside. Unless there are special customization requirements for the radar antenna, it is advisable

to avoid using laminated radomes. The following table presents the characteristics of some commonly used mmWave radar radome materials:

Table 2-1 Material Characteristics of mmWave Radar Radome

Material	Dielectric constant DK	Loss tangent DF
ABS	2.0-3.5	0.0050-0.019
Polytetrafluoroethylene (PTFE)	2.1-2.3	0.0002-0.0003
Polyethylene (PE)	2.3	0.0003
Polycarbonate (PC)	3-3.2	0.005-0.008
Polyimide (PI)	3.4-3.5	0.003-0.005
glass	5.75	0.003
ceramics	9.8	0.0005

3. Example of Radome Design

There are many types of radar radomes, with the more common ones being flat, domed, and hemispherical. However, regardless of the type of structure chosen, it is necessary for the radar radome to have uniform thickness and texture, as well as a smooth surface, to ensure that the loss value experienced by electromagnetic waves passing through each part of the radome at the same angle is consistent, and that the phase delay remains consistent as well.

This time, taking RS6130 as an example, we designed three different types of radar radomes (flat, arched, and hemispherical), using ABS as the material with a dielectric constant of 2.8 and a loss tangent of 0.01.

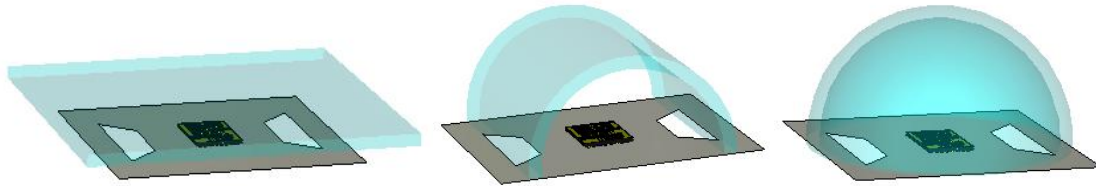
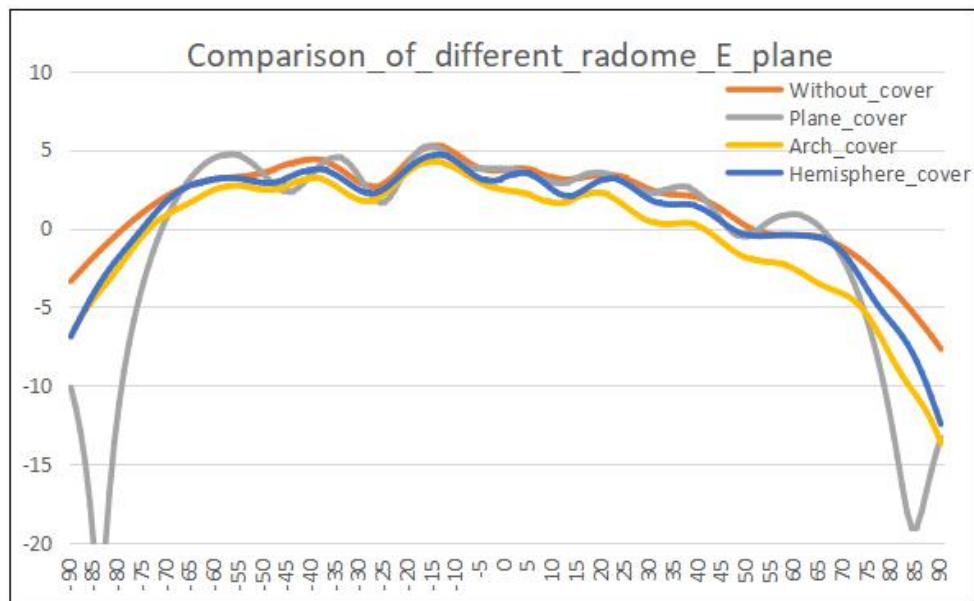


Figure 3-1 Schematic diagrams of three types of radomes

3.1. Comparison of Three Types of Radomes

According to formulas (1) and (3), the wall thickness of the radar radome and the spacing from the inner surface of the radome to the upper surface of the radar can be calculated. The wall thickness T can be taken as 1.5mm, and the spacing can be taken as half a wavelength $D=2.5\text{mm}$.

For the arched and hemispherical covers, their thicknesses can be calculated according to formula (1). Their inner radii can be calculated according to the formula $R=A/2+N*\lambda_0/2$, where A is the antenna aperture and N is a positive integer. The arched cover has a curved surface facing the narrow side of the radar, so the antenna aperture is set to 5.8mm, while the hemispherical cover has no directionality, so the antenna aperture is set to 8.8mm. Finally, the wall thickness of the radome $T=1.5\text{mm}$, the inner radius of the arched cover is set to 10.4mm, and the inner radius of the hemispherical cover is set to 14.4mm. The simulation results of the three types of covers are compared as follows:



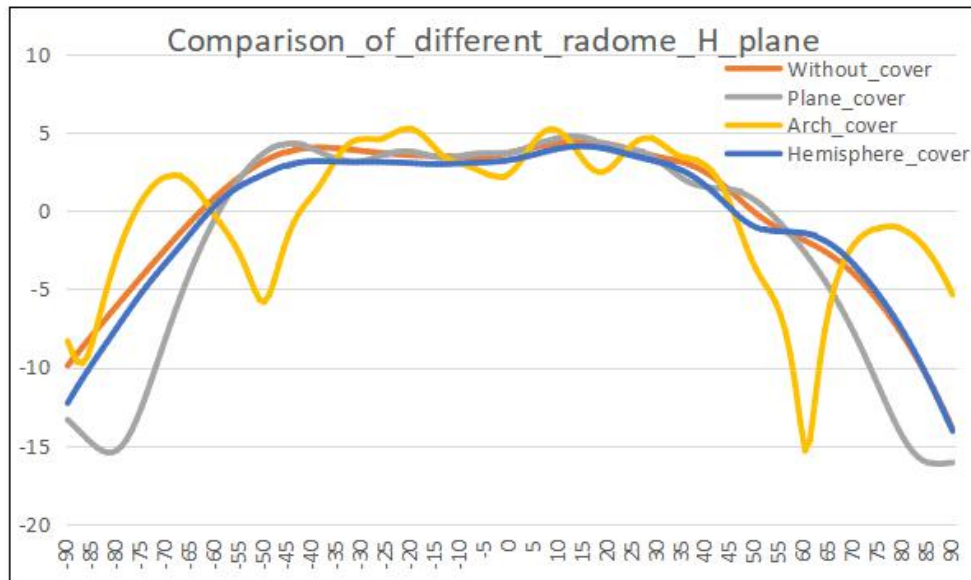


Figure 3-2 Comparison of radiation patterns between three types of radomes and those without radomes

Based on the comparison of radiation patterns above, it can be concluded that, given sufficient size and space, the comprehensive performance of the three types of radomes follows the order: hemispherical > flat ≥ arched. The radiation pattern of the hemispherical radome is closer to that without a radome, making it more "transparent" to the antenna. The beam width of the radiation pattern of the flat radome will be narrowed to some extent, and the ripples in the radiation pattern at certain angles will intensify. The arched radome itself has a certain degree of directionality, with the non-circular surface narrowing the beam width towards the radiation direction, introducing nulls in the pattern, and also deteriorating the angle measurement error to some extent. The above outlines the advantages and disadvantages of different types of radar antenna radomes when there is sufficient radar space.

3.2. Example of Radome Design

In practical radar application scenarios, the size limitations of the radar system may not meet certain spatial requirements. The spacing between the radar radome and the radar can be adjusted to reduce the height of the radar system. When the spacing cannot meet the requirement of one-half wavelength in air, a spacing of at least one-quarter wavelength or more can be selected. Taking RS6130 as an example, the material is still ABS, with a dielectric constant of 2.8 and a loss tangent of 0.01. The wall thickness is chosen to be 1.5mm. The spacing between the top of the radome and the antenna is adjusted by the offset D_{offset} (initial value is 0) of the radome along the antenna radiation direction. For flat radomes, the spacing refers to an integer multiple of one-half wavelength in air, while for arched and hemispherical radomes, it refers to the case where the center of the circle is at the center of the radar surface. For arched and hemispherical radomes, the radius of the radome needs to be adjusted simultaneously to change the curvature and thus adjust the performance of the radome. The adjusted radome results are as follows:

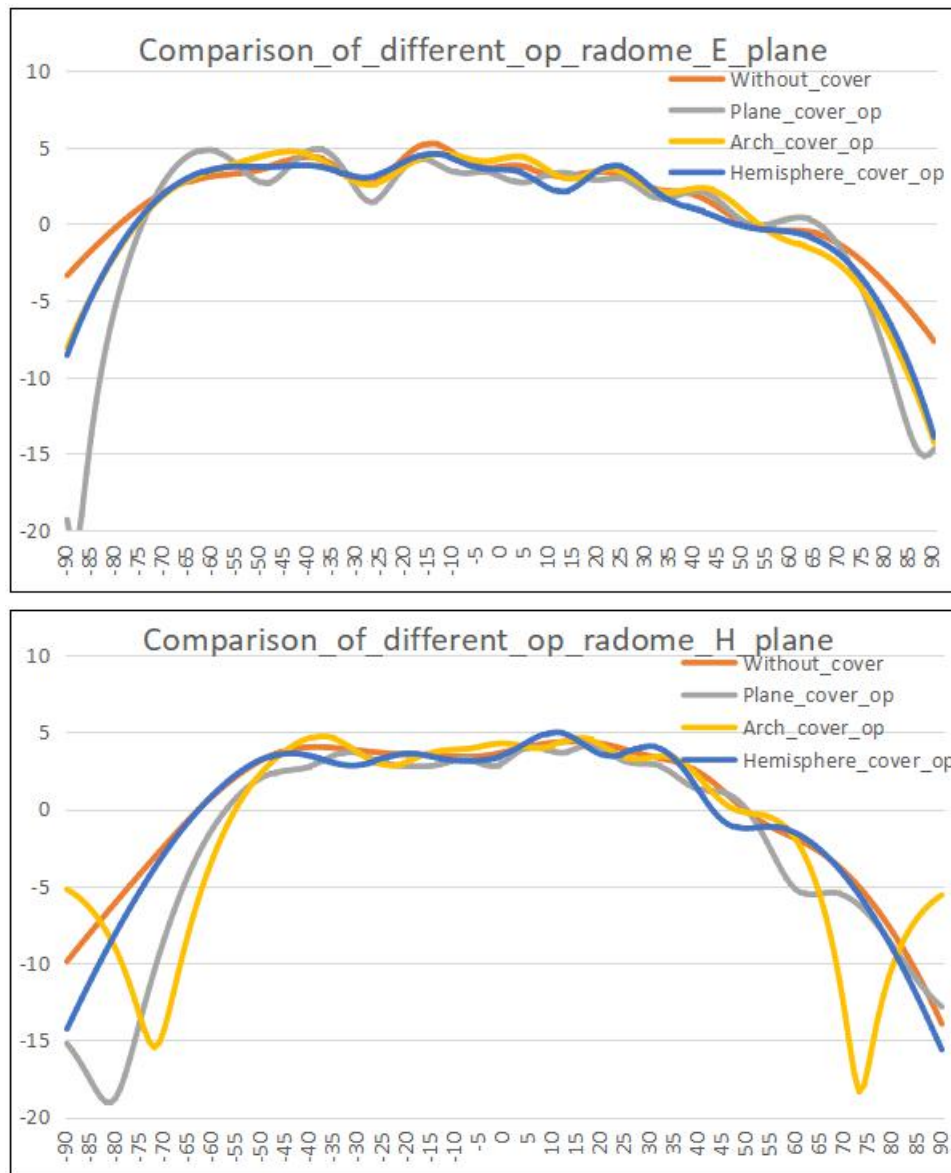


Figure 3-3 Comparison of radiation patterns after adjustment of three types of radomes

It can be seen that, under space constraints, with certain adjustments, the performance requirements of mmWave radar can generally be met. The main dimensions of the radome design in this chapter are shown in the table below (allowable tolerance: $\pm 15\%$):

Table 3-1 Parameter Table of mmWave Radar Radome (Material Selected: ABS)

Type	Wall thickness T (mm)	Distance D (mm)	Inner radius R _{in} (mm)	Outer radius R _{out} (mm)	Radiation direction offset D _{offset} (mm)
Tablet	1.5	2.5	-	-	0
Arch	1.5	-	10.4	11.9	0
Hemisphere	1.5	-	14.4	15.9	0
After adjusting the tablet	1.5	1.25	-	-	-1.25
After arch	1.5	-	20.4	21.9	-15.3

adjustment					
After hemisphere adjustment	1.5	-	21.9	23.4	-16.8

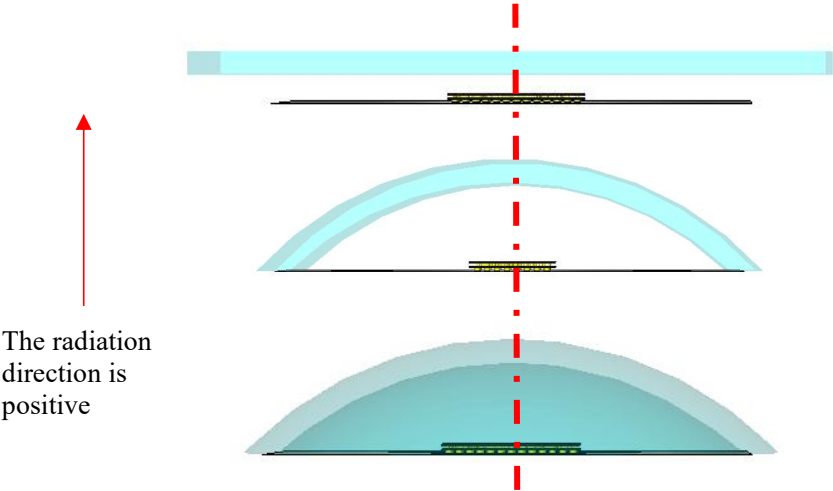


Figure 3-4 Schematic diagram of adjustment for three types of radomes

4. Description of Radar Antenna Clearance Zone

Radar needs to be installed in a suitable environment to avoid the impact of other environmental factors on the antenna performance, which in turn affects the detection performance and accuracy of the radar. For antennas, in addition to the impact of the radar radome itself, the structural components of the radome bracket and the surrounding environment of the antenna may also have a serious impact on antenna performance. Therefore, a good clearance area is required around the antenna.

Taking RS6130 as an example, for different FOV requirements, the antenna radiation area within the FOV coverage needs a good clearance area. This area refers to the radiation near-field area extending outward from the radar body by more than or equal to half the wavelength of the air (2.5mm) at an elevation angle of $(180^\circ - \text{FOV})/2$ or above. Having a good environment in this area can ensure the radiation coverage performance of the antenna, as shown in the figure below:

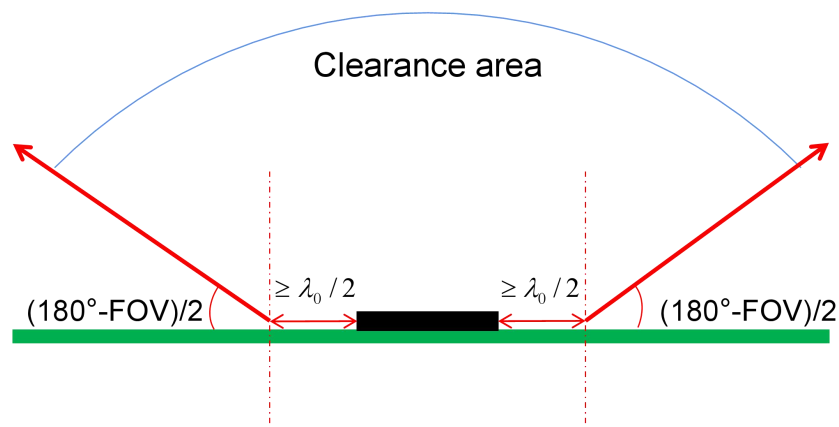


Figure 4-1 Schematic diagram of the clearance area of mmWave radar

5. Optimization of PCB Flooring Performance

In addition to the radome support structure and the radar's surrounding environment, which can affect the radar's performance, the size of the PCB floor also has a certain impact on the radar. Therefore, when reserving PCB dimensions, structural engineers should consider relevant constraints.

For specific details, please refer to Section 8.1 "PCB Board Performance Optimization" in the document: [Data Package\Hardware_Kit\01_Document\RS6xxx_Hardware Design and PCB Layout Guide Manual_Vx.x.pdf](#)

6. Summary

6.1. Summary Description

To achieve a mmWave radar with excellent performance, the design of the radar radome must adhere to relevant design rules for radomes, in order to reduce reflection on the radome surface during signal transmission and reception, and minimize loss and beam distortion.

In addition, for radomes requiring thin-walled design, some studies have also indicated that thin-walled design is more suitable for use in low-frequency situations with relatively large wavelengths, typically with a wall thickness less than or equal to $\frac{\lambda}{10\sqrt{\epsilon_r}}$ 0.01 mm. However, radomes with this type of wall thickness can

only be applied in situations with relatively narrow bandwidths.

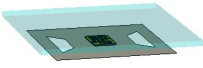
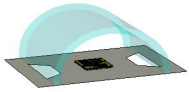
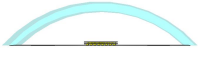
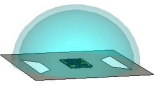
This design guide serves only as a reference for the design of single-layer solid wall type radomes. For multi-layer structure type radomes, their characteristics are more complex compared to single-layer structures, and require separate analysis and design accordingly.

In addition, our company has provided a 3D model, which includes an example of radome design and a schematic of FOV.

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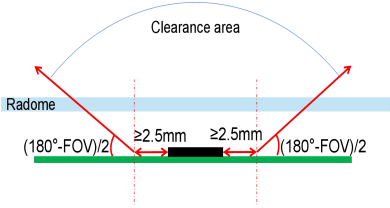
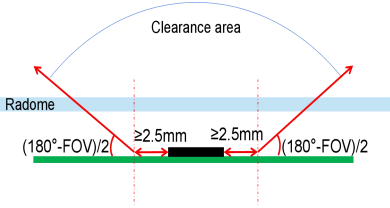
6.2. Summary of Reference Designs

Table 6-1 Reference design of mmWave radar radome

Chip Type	RS6130	RS6240	Parameter Description
 Flat cover	T=1.5mm D≥1.25mm	T=1.5mm D≥1.25mm	T: Cover thickness The reference design T must meet the following requirements: $T=\lambda_m/2$ or $T=\lambda_m/10$ $\lambda_m=c/(f_0*\sqrt{\epsilon_r})$ That is, the thickness of the cover satisfies 1/2 of the wavelength of the medium or 1/10 of the wavelength of the medium Material ABS: $\epsilon_r=2.8$ D: Distance between the inner wall of the cover and the upper surface of the chip
 Arched cover 1	T=1.5mm R_in=10.4mm R_out=11.9mm	T=1.5mm R_in=11.9mm R_out=13.4mm	D_offset: The offset along the radiation direction between the inner wall of the cover and the upper surface of the chip, with positive values towards the radiation direction and an initial value of 0. For flat covers, it refers to the case where the spacing is an integer multiple of one-half of the wavelength of air, while for arched and hemispherical covers, it refers to the case where the
 Arched cover 2	T=1.5mm R_in=20.4mm R_out=21.9mm D_offset=-15.3mm	T=1.5mm R_in=21.9mm R_out=23.4mm D_offset=-16.9mm	
 Hemispherical cover 1	T=1.5mm R_in=14.4mm R_out=15.9mm	T=1.5mm R_in=16.2mm R_out=17.7mm	

 Hemispherical cover 2	T=1.5mm R_in=21.9mm R_out=23.4mm D_offset=-16.8mm	T=1.5mm R_in=23.7mm R_out=25.2mm D_offset=-18.7mm	center of the circle is at the center of the radar surface. R_in: Inner radius of the cover R_out: Outer radius of the cover The tolerance for the aforementioned dimensions is ±15%
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Table 6-2 Reference design for clearance area of mmWave radar

Chip	RS6130	RS6240
Schematic diagram of structural environment		

6.3. Self-Inspection Checklist

Table 6-3 Self-Inspection Checklist

Serial Number	Check Item
1	The recommended radome shape mentioned above must be adopted. If a non-recommended shape is used, performance cannot be guaranteed, and it is prone to cause distortion in the directional pattern and deterioration in angle measurement accuracy, thereby resulting in radar detection performance that does not meet expectations.
2	The thickness of the radome must adhere to the recommended parameters outlined in the reference design summary in Section 6.2 . If the material of the radome differs significantly from the material with the recommended dielectric constant in the reference design summary in Section 6.2 , the thickness of the radome should be calculated based on the actual dielectric constant of the material.
3	It is necessary to confirm that the surrounding structure of the antenna does not infringe upon the field of view (FOV) of the radar, with the starting position of the FOV located at a distance of 2.5mm from the radar chip.
4	The distance from the antenna surface to the radome meets the recommended parameters outlined in the reference design summary in Section 6.2 .
5	The reserved PCB size and shape must meet the constraints outlined in Section 5, PCB Flooring Performance Optimization .

6.4. Q & A

Question 1: Is it feasible for the radome to be constructed by stacking multiple media?

Answer: If the dielectric constants of the media are the same or similar (within 15%), it is acceptable. However, there can be no more than two types of media, and the thickness of the adhesive between the media should not exceed 0.5mm. The total thickness after stacking must still meet the aforementioned design requirements for the thickness of the radome.

Question 2: What if the structural design cannot fully meet the constraint requirements?

Answer: If there are requirements for angle measurement and low power consumption over long distances, the design must strictly adhere to the constraints. If structural limitations prevent adjustments, it can lead to uncontrollable deterioration in detection performance. This can be confirmed through prototyping and actual performance testing, or, if conditions and professional expertise permit, through simulation.

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